

Averting Obesity and Type 2 Diabetes in India through Sugar-Sweetened Beverage Taxation: An Economic-Epidemiologic Modeling Study

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Evidence abstract

BACKGROUND Taxing sugar-sweetened beverages (SSBs) has been proposed in high-income countries to reduce obesity and type 2 diabetes.

We sought to estimate the potential health effects of such a fiscal strategy in the middle-income country of India, where there is heterogeneity in SSB consumption, patterns of substitution between SSBs and other beverages after tax increases, and vast differences in chronic disease risk within the population.

METHODS AND FINDINGS Using consumption and price variations data from a nationally representative survey of 100,855 Indian households, we first calculated how changes in SSB price alter per capita consumption of SSBs and substitution with other beverages.

We then incorporated SSB sales trends, body mass index (BMI), and diabetes incidence data stratified by age, sex, income, and urban/rural residence into a validated microsimulation of caloric consumption, glycemic load, overweight/obesity prevalence, and type 2 diabetes incidence among Indian subpopulations facing a 20% SSB excise tax.

The 20% SSB tax was anticipated to reduce overweight and obesity prevalence by 3.0% (95% CI 1.6%-5.9%) and type 2 diabetes incidence by 1.6% (95% CI 1.2%-1.9%) among various Indian subpopulations over the period 2014-2023, if SSB consumption continued to increase linearly in accordance with secular trends.

However, acceleration in SSB consumption trends consistent with industry marketing models would be expected to increase the impact efficacy of taxation, averting 4.2% of prevalent overweight/obesity (95% CI 2.5-10.0%) and 2.5% (95% CI 1.0-2.8%) of incident type 2 diabetes from 2014-2023.

Given current consumption and BMI distributions, our results suggest the largest relative effect would be expected among young rural men, refuting our a priori hypothesis that urban populations would be isolated beneficiaries of SSB taxation.

Key limitations of this estimation approach include the assumption that consumer expenditure behavior from prior years, captured in price elasticities, will reflect future behavior among consumers, and potential underreporting of consumption in dietary recall data used to inform our calculations.

CONCLUSION Sustained SSB taxation at a high tax rate could mitigate rising obesity and type 2 diabetes in India among both urban and rural subpopulations.

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